



WISDOM BETTER KNOWLEDGE

**Faculty of Science and Technology**  
**Jagran School of Computer Applications**

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Assignment : 3

Q1 Solve the following using distance vector algorithm for the shortest path.

(i) Graph with A, B, C, D.

Edges :  $A - B = 8$

$B - C = 2$

$C - D = 3$

$D - A = 5$

→ from A :

- $A \rightarrow B = 8$

- $A \rightarrow D = 5$

- $A \rightarrow C$

- $A \rightarrow B \rightarrow C = 8 + 2 = 10$

- $A \rightarrow D \rightarrow C = 5 + 3 = 8$  [Shortest]

Final shortest paths from A.

$A \rightarrow B = 8$

$A \rightarrow C = 8$

$A \rightarrow D = 5$

→ from B :

- $B \rightarrow A = 8$

- $B \rightarrow C = 2$

- $B \rightarrow D$

- $B \rightarrow C \rightarrow D = 2 + 3 = 5$  (Shortest)

- $B \rightarrow A \rightarrow D = 8 + 5 = 13$



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(ii) Graph with 0, 1, 2, 3

Edges:  $0-1 = 1$

$$1-2 = 1$$

$$2-3 = 2$$

$$0-3 = 7$$

$$0-2 = 3$$

from 0 to

$$0 \rightarrow 1 = 1$$

$$0 \rightarrow 2 :$$

$$\text{by direct} = 3$$

$$\text{via } 2 = 1 + 1 = 2 \text{ (shortest)}$$

$$0 \rightarrow 3 :$$

$$\text{by direct} = 7$$

$$\text{via } 2 = 2 + 2 = 4$$

final :

$$0 \rightarrow 1 = 1$$

$$0 \rightarrow 2 = 2$$

$$0 \rightarrow 3 = 4$$

from 1 to

$$1 \rightarrow 0 = 1$$

$$1 \rightarrow 2 = 1$$

$$1 \rightarrow 3 = 2 + 1 = 3$$

from 2 to

$$2 \rightarrow 1 = 1$$

$$2 \rightarrow 3 = 2$$

$$2 \rightarrow 0 = 2 + 1 = 3$$

from 3 to

$$3 \rightarrow 2 = 2$$

$$3 \rightarrow 1 = 2 + 1 = 3$$

$$3 \rightarrow 0 = 2 + 1 + 1 = 4$$



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## Q2 Protocols Explanation

(i) HTTP (Hypertext Transfer Protocol):

- used for web communication
- works for port 80
- stateless protocol
- Transfer HTML pages

(ii) HTTPS:

- Secure version of HTTP
- Uses SSL / TLS encryption
- Port 443
- Protects data from hackers

(iii) FTP (File Transfer Protocol):

- Transfer file between client & server
- Ports 20 & 21
- Not secure (plain text transfer)

(iv) TCP (Transmission Control Protocol)

- Connection-oriented
- Reliable
- Ensure ordered delivery

(v) UDP (User Datagram Protocol)

- Connectionless
- Faster but unreliable
- used in video streaming, gaming.



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(vi) DNS (Domain Name System)

- Converts domain → IP address
- Ex: google.com → 14. x. x. x.
- works on port 53

(vii) RIP (Routing Information Protocol)

- Distance vector routing protocol
- Uses hop count.
- Max hops = 15
- Slow convergence.

Q3 What is Congestion Control in computer networks?  
Explain leaky bucket algorithm in detail.

Ans Congestion Control: It prevents network overload when too much data is sent.

Problems: Packet loss

Delay

Reduced Performance.

→ Leaky Bucket algorithm:

Idea: Data flows like water from a bucket at a constant rate.

- Working:
  - Incoming packets stored in a bucket [queue]
  - Packets sent at fixed rate
  - If bucket overflows → packets dropped.



Advantage : Smooth traffic flow  
Prevents congestion

Disadvantages : Doesn't handle heavy traffic well.

Q. Explain all the protocols used in email for sending & receiving message in detail.

Ans)

(i) SMTP (Simple mail transfer Protocol)

↳ used to send emails

↳ Port 25/587

↳ works between mail servers.

(ii) POP 3 (Post office Protocol v3)

↳ used to receive emails

↳ Download emails to device

↳ Deletes from server.

(iii) IMAP (Internet Message Access Protocol)

↳ Also used to receive emails

↳ keeps email on server.

↳ Sync. across devices

Q. Flow of Email:

1. Sender uses SMTP

2. Mail server processes

3. Receiver uses POP 3 / IMAP.



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